

North American Particle Accelerator Conference (NAPAC 2019)

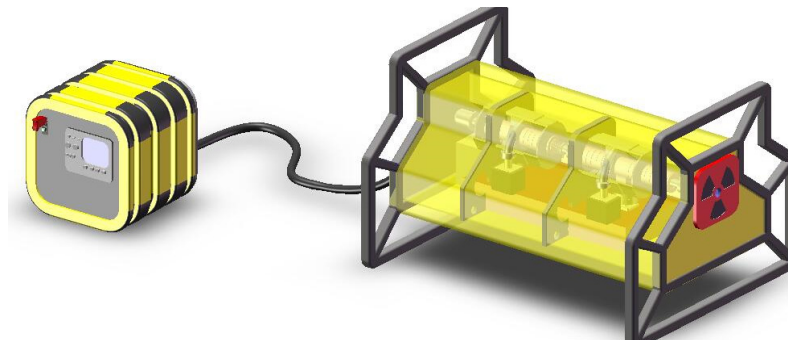
Ultra-Compact Accelerator for Radioactive Isotope Sources Replacement, Security, NDT and Medical Applications*

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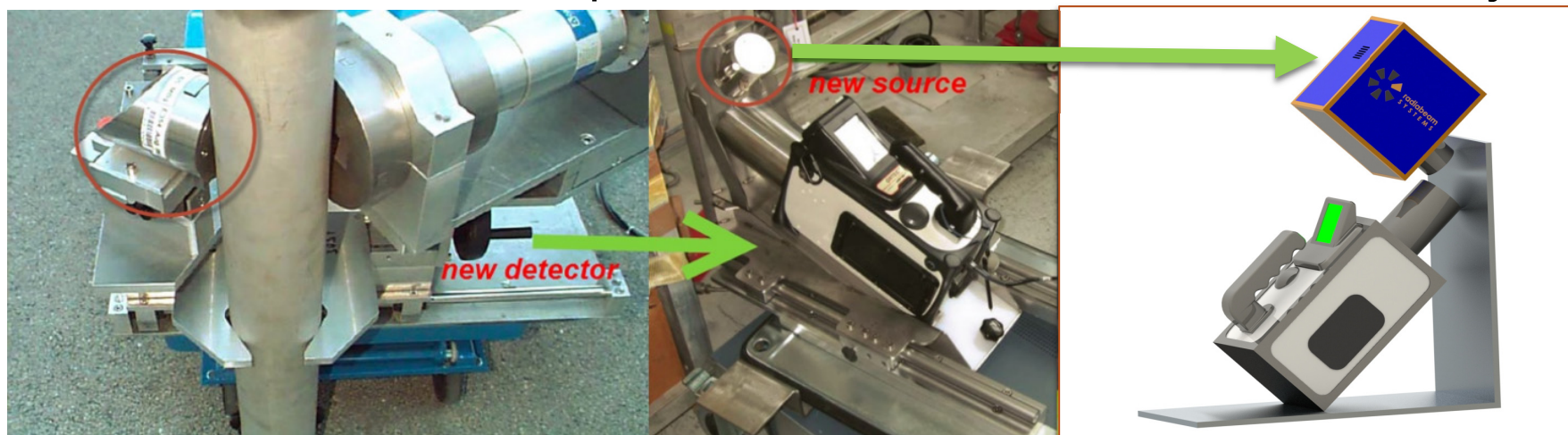
September 4, 2019

- IAEA, DOE and the DHS have identified as a priority the replacement of radioactive sources
 - Between 1993 and 2006, there were 1,080 incidents of unauthorized activities with radiological materials worldwide.
 - 80% of the radionuclide devices in the US are Category 2 gamma emitters (Ir-192, Cs-137, Co-60 and Se-75),
 - Used for industrial radiography, well logging, or in self-contained irradiators (e.g., blood irradiators).
 - Radioactive sources can be replaced with particle accelerators
 - Accelerators must be reasonably compact and inexpensive



Co-57-based safeguard tools

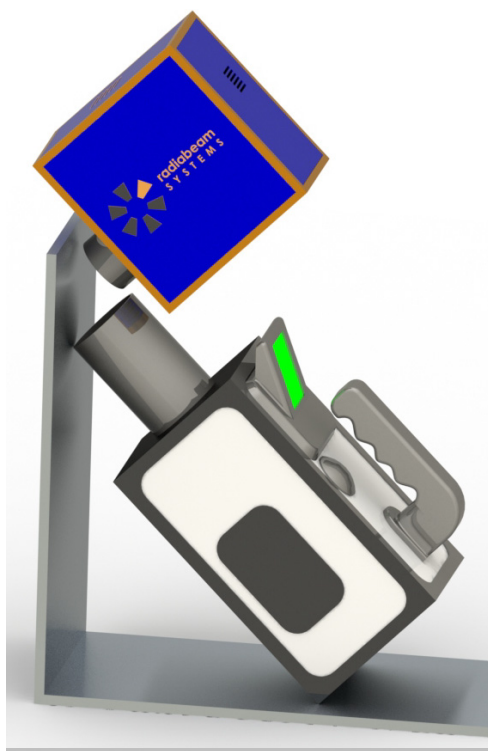
- One of the key instruments for safeguards at enrichment plants is the Cascade Header Enrichment Monitor (CHEM)
- The detector is used to detect the presence of UF_6 gas at low pressures and to determine whether this UF_6 is enriched
 - UF_6 irradiation with γ -rays above the uranium K-edge energy
 - Measuring both the passive emission of the uranium at 185 keV and its induced fluorescence at 98 keV
- Current Co-57 detector produces several mCi of 122 keV X-rays



* Cascade Header Enrichment Monitor. More details in A. Lebrun et al., "Improved verification methods for safeguards verifications at enrichment plants"

Why not X-ray tube?

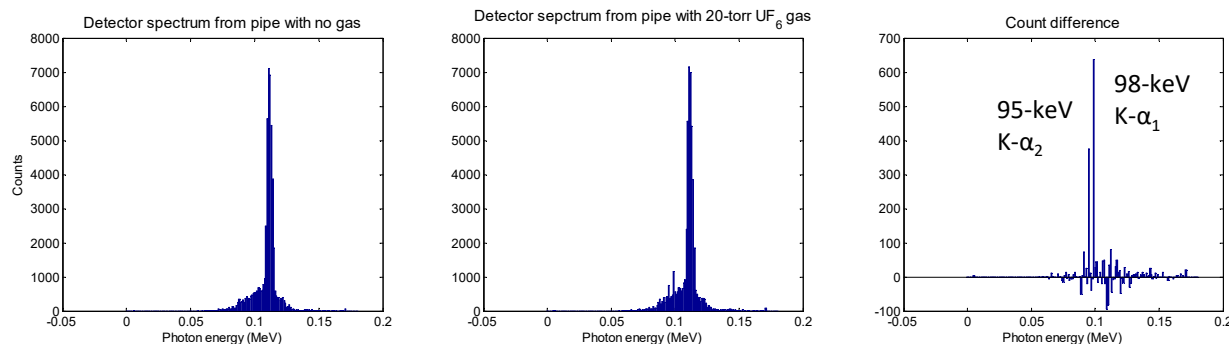
- High voltage X-ray tubes are
 - Bulky
 - Heavy
 - Not practical after 0.5 MeV energies



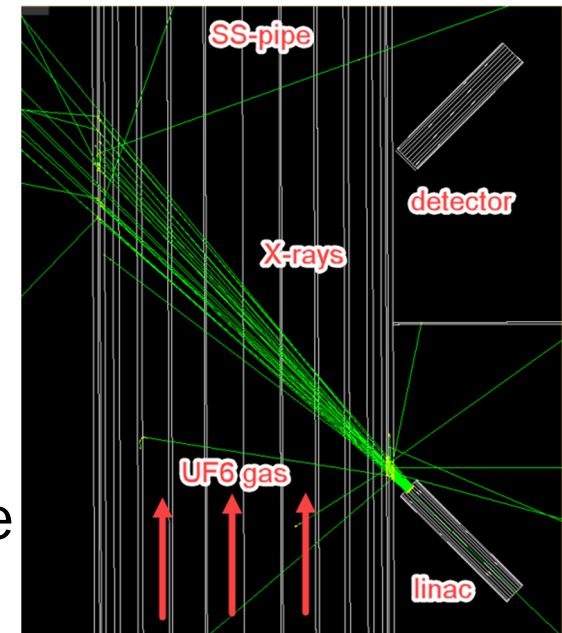
Source	Energy	Size	Weight
GE	200	84x29x27	31
Spellman	200	67x40x17	67
RadiaBeam	180	20x18x10	5-10
	keV	cm ³	kg

Can linac replace Co-57 source?

- Monte-Carlo simulations of the linac, and the detector attached to the pipe with UF_6 gas
 - Thick converter (3mm) to filter low-energies
 - One photon detection per electron bunch to prevent pileup



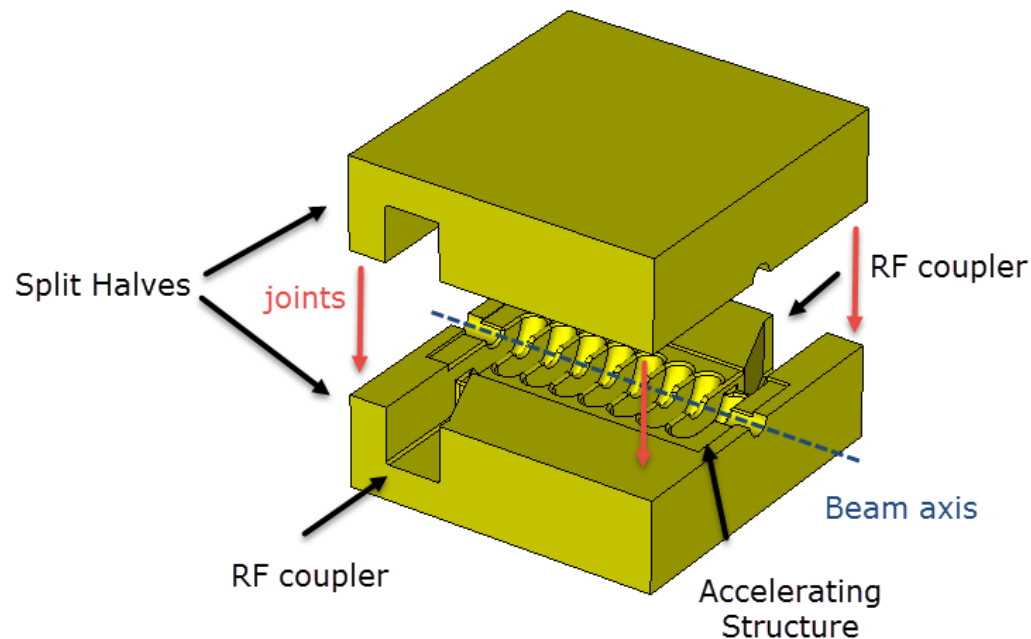
- Replacement with linac is feasible
 - With a proper design and parameters choice



- The simulations defined the requirements for the linac

Regime	Min	High	
Beam energy, keV	180	180	Should not exceed this value not to mask 185 keV fission decay peak
Beam current (peak), mA	2	10+	Depends on the repetition rate and the convertor design
Pulse repetition rate, pps	400	2000	Typical for Ku-band magnetrons
Pulse width, μ s	2.5	0.5	Scales with the repetition rate. Duty factor is $\sim 0.1\%$
Equivalent dose to Co-57 source, mCi	3.6	18	Scales linearly with pulse repetition rate
Scan time, mins	2	~ 30 s	To produce required statistics

- We developed Ku-band (compact) split-structure (low cost) linac as an X-ray source for CHEM detector.
 - Split structure consists of fewer parts
 - Split structure doesn't require tuning



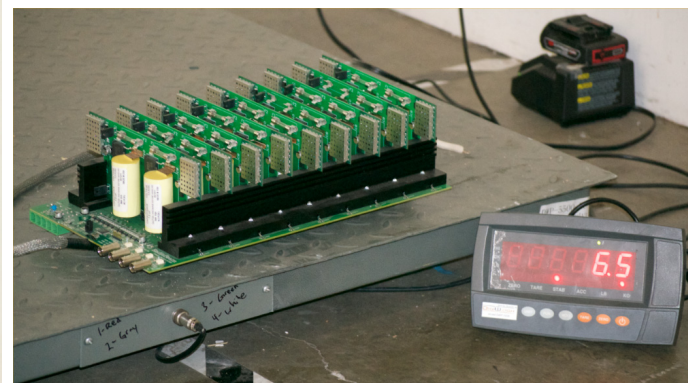
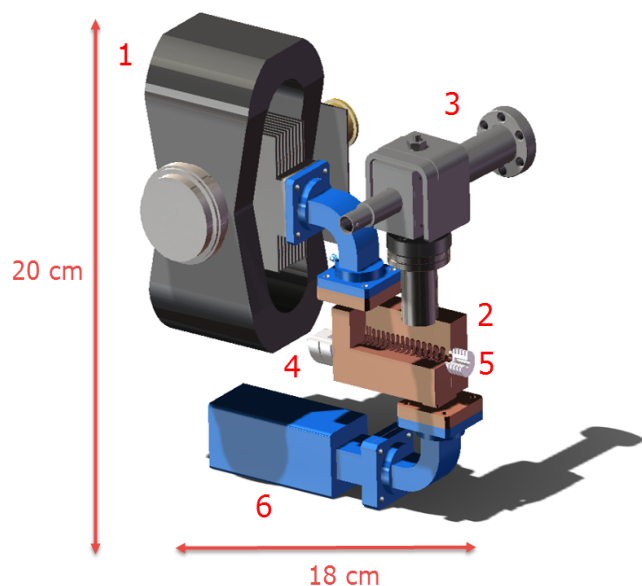
S. Kutsaev et al. Split Structure Particle Accelerators. US Patent Application No. 62/513911, filed June 1, 2017

S. Kutsaev et al. Modified Split Structure Particle Accelerators. US Patent Application No. 62/734971, filed September 21, 2018

- Required maximum energy = 180 keV
- With 2 mA peak current, beam loading is ~400W
- Estimated linac power for ~6 cm linac: 40 kW
- Ku-band is optimal
 - Ka-band can be explored in future

Band	Frequency, GHz	Peak Power, kW	Anode Voltage, kV	Weight, kg	Size, cm ³
S-	2.7 – 2.9	500	31	43	35 x 35 x 32
C-	5.4 – 5.9	320	26	16.5	23.5 x 22 x 22
X-	9.0 – 10.0	130	18	6.5	20 x 16.5 x 11
Ku-	15.6 – 16.7	60	15	3.0	15 x 14 x 9
Ka-	34.5 – 35.5	40	13.5	~2.2	~12 x 11 x 7

- All sub-components are chosen to provide the compactness, be light-weighted and cheap

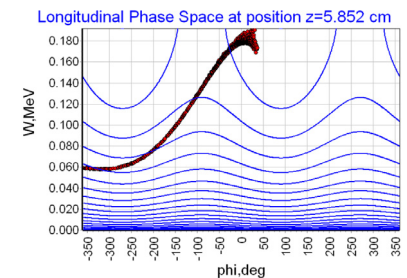
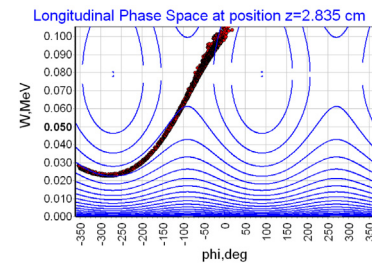
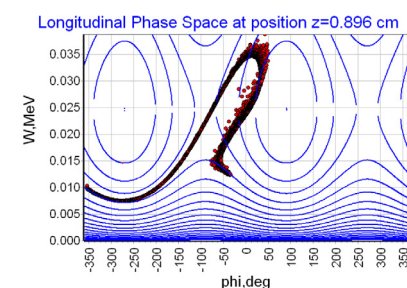
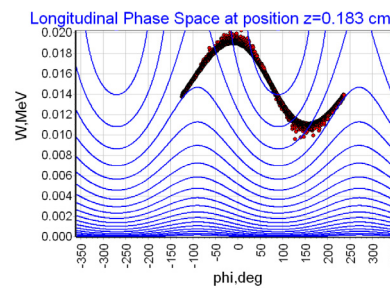
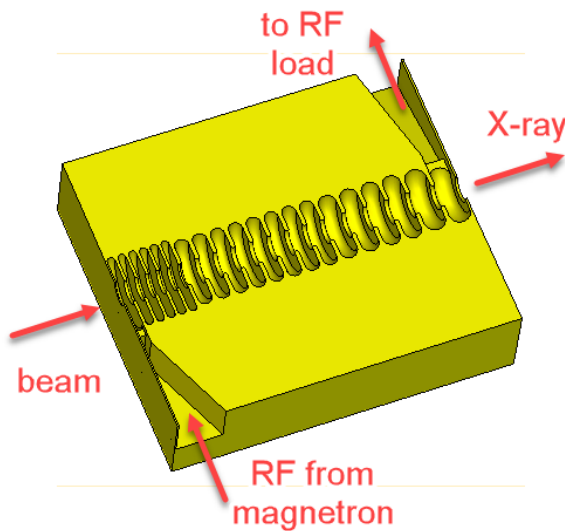


1- magnetron, 2 – split accelerating structure, 3 – vacuum pump, 4 – electron gun, 5 – C-ray converter, 6 – RF load.

Modulator in a separate box

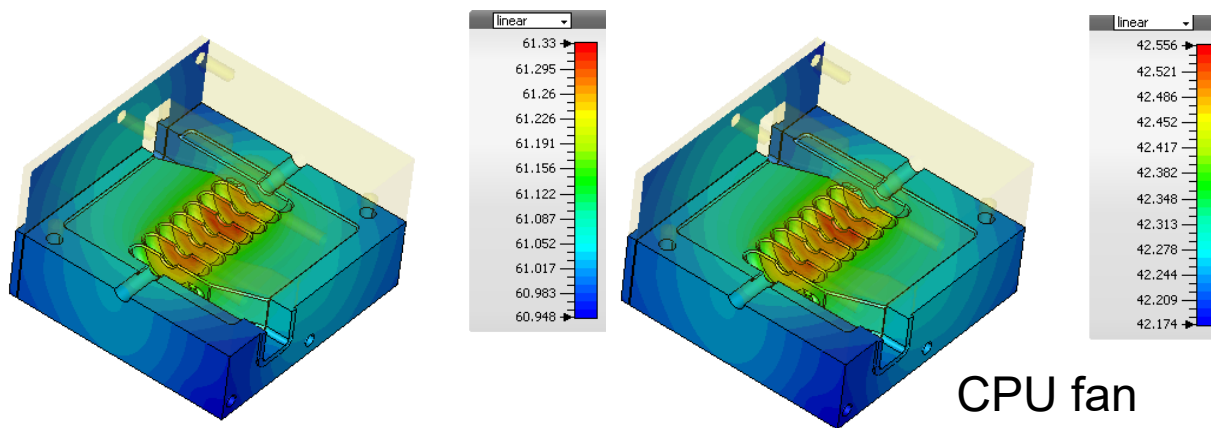
RF structure

- We use $2\pi/3$ TW split-structure
 - Circulator is not required
 - $\pi/2$ -mode SW structure has too small details to machine



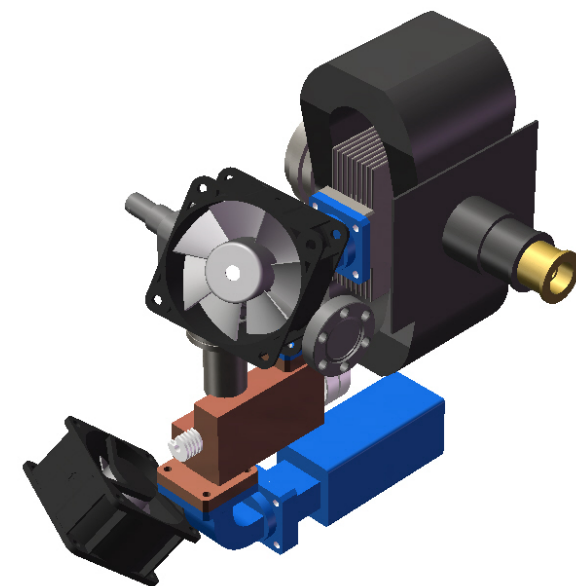
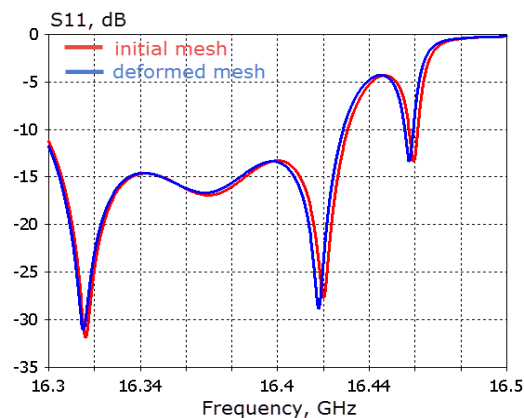
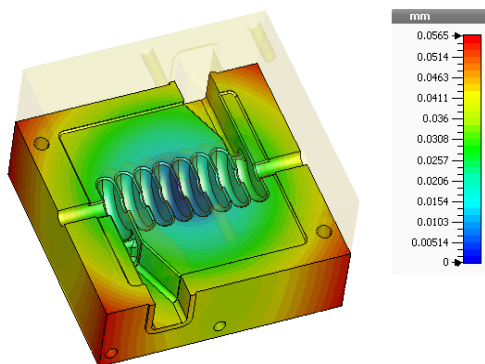
- Transmission is 25% with no solenoid
- Length = 5.9 cm
- Output energy is tunable using magnetron power

- Thermal gradients cause insignificant mechanical stresses and deformations

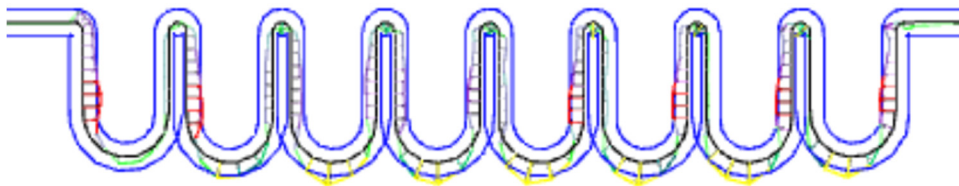
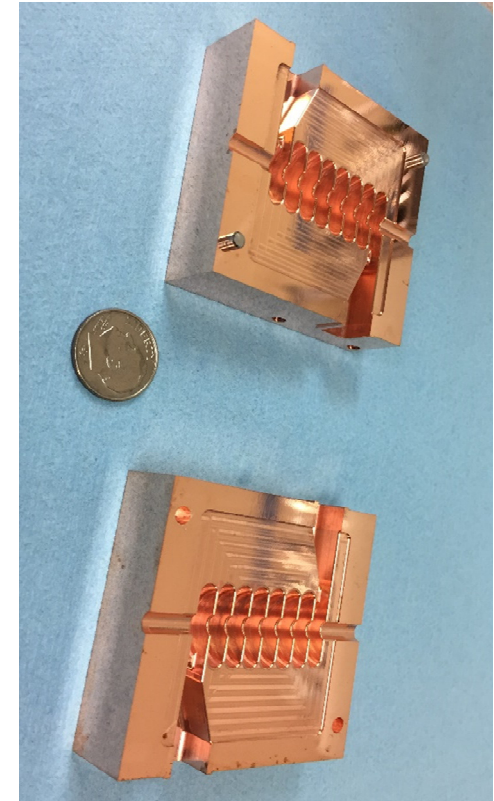
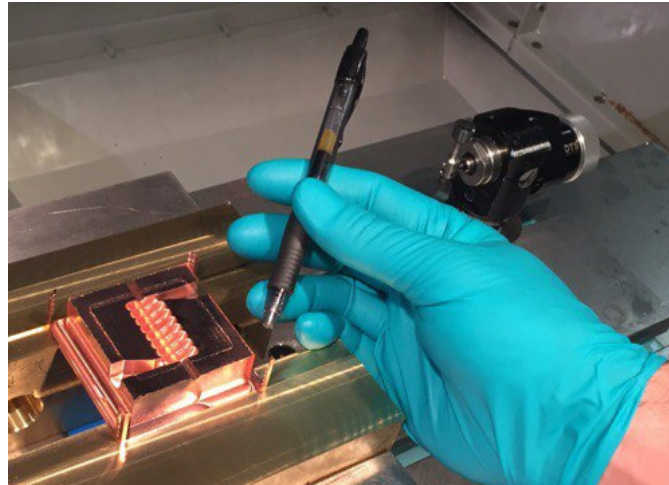
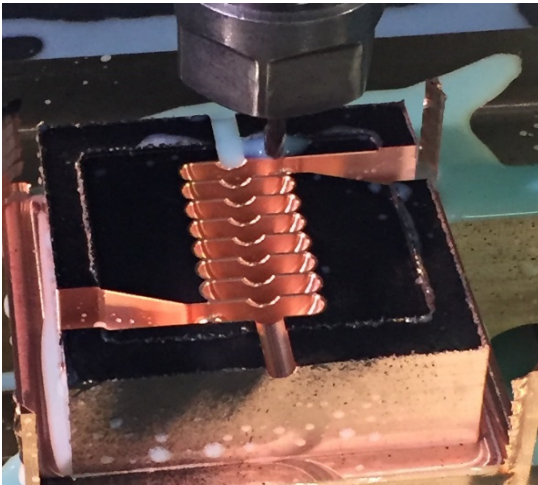


No cooling

CPU fan

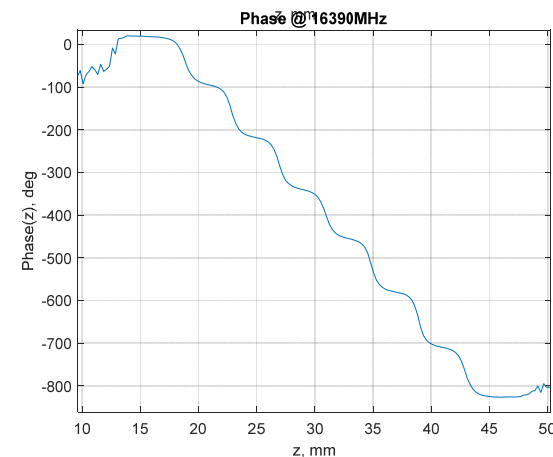
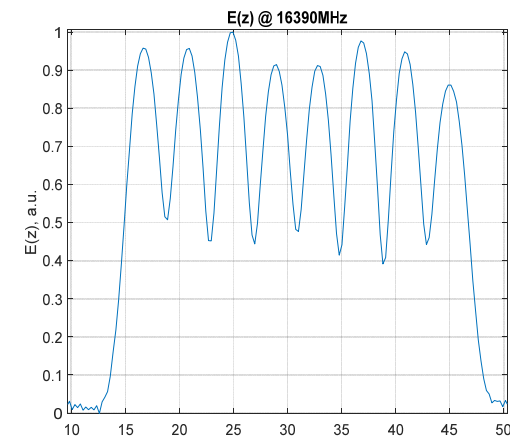
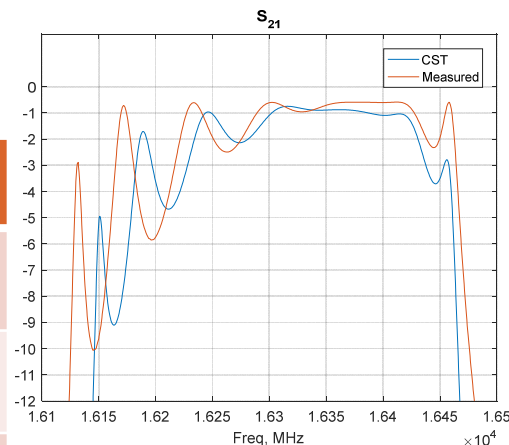
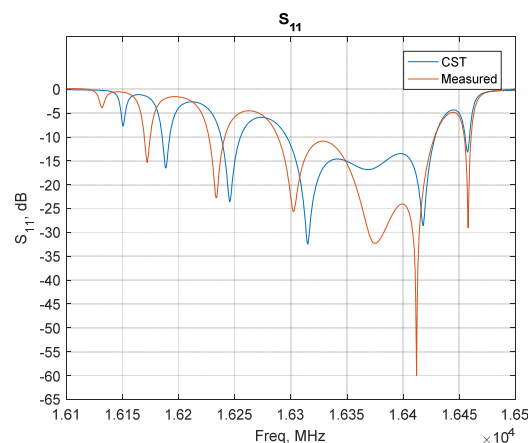
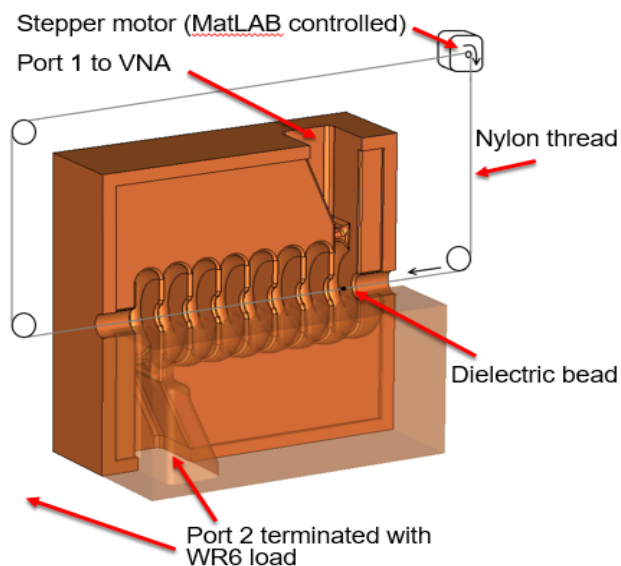


- The structure prototype was milled at RadiaBeam
 - Turned out to be very fast and cost effective
 - EDM as an alternative can be used



Acceptable tolerances: $\pm 20 \mu\text{m}$

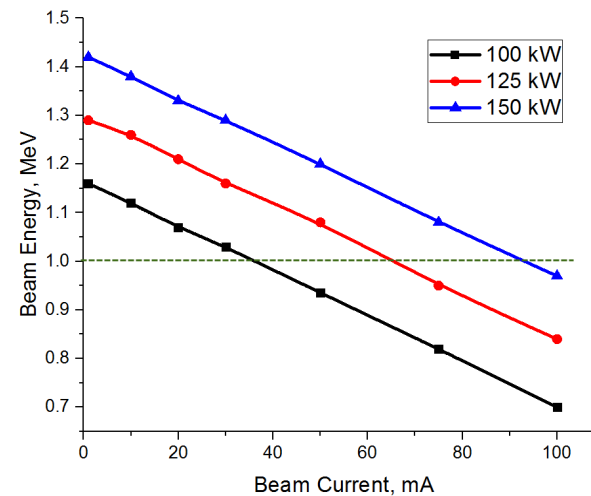
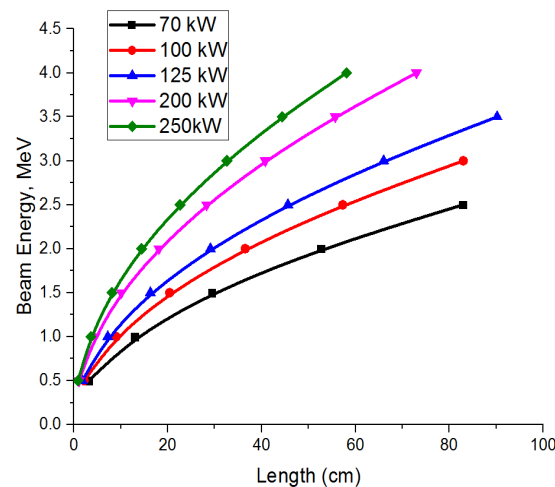
- Excellent agreement between measurements and simulations



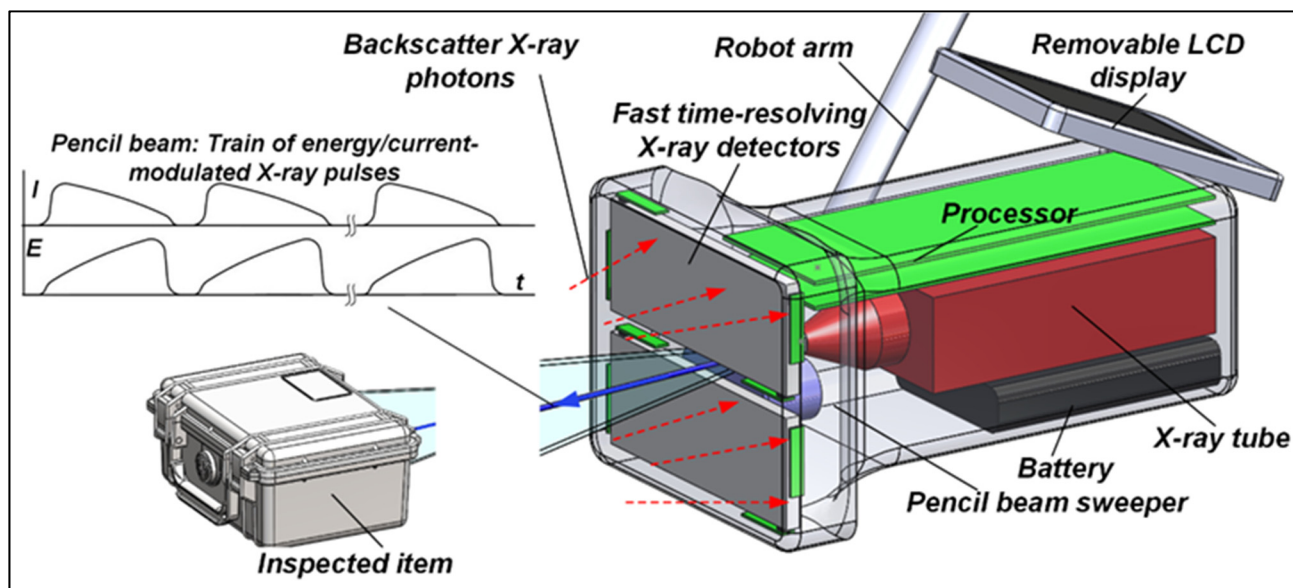
Parameter	Simulated	Measured
Frequency of $2\pi/3$, GHz	16.4	16.395
Group velocity, %	1.4	1.3
Q-factor	6500	6450

- No tuning is needed!

- Category 2 radioactive isotopes replacement
 - Ir-192 (0.38 MeV), Cs-137 (0.66 MeV), Co-60 (1.25 MeV) and Se-75 (0.265 MeV)
 - Higher energy accelerator design is required
 - Higher currents are required
- The design is scalable up to 4 MeV with 200+ kW magnetron
 - At 1 MeV beam energy, average beam power can reach 100 W



- Novel adaptive scanning techniques*
 - CT scan systems for baggage and packages
 - X-ray backscatter imagers
 - Medical CT
 - Requires energy ramp



*A. Arodzero et al. "Adaptive Computed Tomography with modulated-energy X-ray pulses". Provisional Patent App. No. 62/575,403
A. Arodzero et al. "X-ray Backscatter Inspection System". US Patent Application 2017/0338545

- We are building the 1 MeV demonstration prototype
 - 135 kW Ku-band magnetron
 - With energy ramping capability
 - First beam is expected by the end of 2019



Thank you!

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